

Quiz — 1.1.3 Input, Output and Storage Devices

Name: _____ Date: _ **Mark:** ____ / 25

OCR H446 · Component 01 · 1.1.3

Section A — Multiple choice (8 marks)

1. Which of the following is an **output** device? A. Barcode scanner B. Microphone C. Actuator D. Capacitive touchscreen (used to type) Your answer: ☐
2. A **magnetic hard disk drive (HDD)** stores data by: A. Trapping electrons in cells with no moving parts B. Magnetising patterns/polarity on a spinning platter, read by a R/W head C. Using a laser to detect pits and lands D. Holding data only while powered Your answer: ☐
3. Which storage type uses a **laser to detect pits and lands**? A. Magnetic B. Flash / SSD C. Optical (CD/DVD/Blu-ray) D. RAM Your answer: ☐
4. A key advantage of **flash/SSD** storage over magnetic HDDs is that it: A. Has the lowest cost per gigabyte B. Has no moving parts, so it is faster, silent and more durable C. Has unlimited write cycles D. Has the largest possible capacity Your answer: ☐
5. Which statement correctly describes **RAM**? A. Non-volatile and normally read-only B. Volatile, can be read from and written to, holds running programs/data C. Holds the bootstrap/BIOS firmware permanently D. A type of secondary storage Your answer: ☐
6. **ROM** is typically used to store: A. The currently running programs B. Documents the user is editing C. The bootstrap loader / BIOS firmware D. Temporary clipboard data Your answer: ☐
7. **Virtual memory** is best described as: A. Remote storage accessed over the internet B. Cache memory inside the CPU C. The use of secondary storage as extra RAM when physical RAM is full D. A faster type of RAM Your answer: ☐
8. Magnetic **tape** is described as a **serial access** medium, making it most suitable for: A. Random access of individual records very quickly B. Cheap, high-capacity archive/backup C. Running an operating system D. High-speed gaming storage Your answer: ☐

Section B — Short answer (12 marks)

9. State **two** differences between **RAM** and **ROM**. [2]

10. Explain why **secondary storage** is described as **non-volatile**, and give **one** example of a secondary storage device. [2]

11. A user needs portable storage to carry large video files between sites, frequently rewriting them, with fast transfer and good durability. Recommend a suitable storage type and **justify** your choice against the requirements. **[3]**

12. Explain how **virtual memory** works and state **one** drawback of relying on it heavily. **[2]**

13. Define the term **disk thrashing**. **[1]**

14. State **two** advantages of using **virtual (cloud) storage** instead of local storage. **[2]**

Section C — Exam-style question (5 marks)

15. A busy office needs a printer to produce a high volume of black-and-white text documents quickly and cheaply each day. A separate home user occasionally prints a few colour photographs.

Recommend a suitable type of printer for **each** user and justify your choices by comparing the two printer types against the stated requirements. **(AO2/AO3) [5]**

Office: ____

Home: ____
